Test Plan Document — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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Version: 3.1 · **Date:** 2026-11-01 (3.1 adds IT_038 for NFR-012)
Status: Test design complete; execution during Sprint 1 (19–23 Oct) and Sprint 2 (26–30 Oct)

1. Purpose

Test cases for the Online Examination Portal, written to the template issued in `Test Plan Document.docx`. Coverage is **8–10 test cases per use case** across **13 use cases**, spanning **Unit, Integration and System** testing, and collectively exercising **all 30 functional requirements** in `Documents/SRS.pdf`. Requirement IDs are those of the SRS; use case IDs are those of `Documents/Use_Case_Diagram.pdf` and `Documents/Use_Case_Flow.pdf`.

2. Test Case ID convention

Per the template's guidance (*"follow some convention like `UT_01`"*):

Prefix	Test type	What it exercises
UT_nnn	**Unit test**	One function, validator or rule in isolation — field validation, a scoring calculation, a state transition
IT_nnn	**Integration test**	Two or more components together — API to database, service to mailer, UI to API
ST_nnn	**System test**	An end-to-end flow through the running system as a real actor would drive it

Numbering is a single running sequence per prefix, so an ID is never reused if a case is later inserted or retired.

3. Traceability

Every test case description opens with the requirement ID it verifies, so this document feeds the test-case columns of the Requirements Traceability Matrix without a separate mapping table. Identifiers are the FR-nnn IDs of the SRS, which are also the backlog items on the GitHub Project.

4. Testing environment

Environment name	TEST — dedicated test instance, seeded database, isolated from demo data
Client	Google Chrome (latest stable) on Windows 11, 1920×1080; a second browser profile where a test needs two concurrent sessions
Server	Application server with an NTP-synchronised system clock
Mail	Captured SMTP sink, so system mail is readable without a live mailbox
Data	Seeded fixture set — see §5

5. Standing test fixtures

Referenced by name in the *Test data* column so precise values are not repeated in every row.

Fixture	Value
STU_A	stu.a@pesu.pes.edu / Passw0rd!A / roll PES1UG24CS001, role Student, active
STU_B	stu.b@pesu.pes.edu / Passw0rd!B / roll PES1UG24CS002, role Student, active
FAC_A	fac.a@pes.edu / Fac#2026A, role Faculty, assigned to CRS_X
FAC_B	fac.b@pes.edu / Fac#2026B, role Faculty, assigned to CRS_Y only
ADM_A	admin@pes.edu / Adm#2026A, role Administrator
CRS_X	Course UE24CS351A / "Database Management Systems" / Semester 5, active
CRS_Y	Course UE24CS342AA5 / "Advanced Virtual Reality" / Semester 5, active
EXM_1	Examination "DBMS Unit Test 1", CRS_X, 60 min, declared total 50, 25 questions, published
Q_SC1	Single-correct MCQ, 4 options, correct = option B, +2 marks, −0.5 negative

6. How the result columns are used

Actual Result and *Test Result* are deliberately **left blank**. The template requires them to be populated from manual execution, and no implementation exists yet — Sprint 1 starts on 19 Oct. Filling them now would be fabrication. Each row is executed and populated in the sprint that implements its requirement (see the Sprint column of the SRS), and the results are committed with that sprint's pull requests.

UC-01 — Register a Student account

Verifies FR-001.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_001	User Registration	FR-001 — Register with	Registration page	1: Open /register	Name Arjun Rao, email	Account created with		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		all four mandatory fields valid	reachable; no account exists for the email or roll number	2: Enter full name, institutional email, roll number, password 3: Click Register	arjun.rao@pesu.pes.edu, roll PES1UG24CS111, password Str0ng#Pass	role Student; message "Registration successful"		
UT_002	User Registration	FR-001 — Reject submission with a mandatory field left empty	Registration page reachable	1: Open /register 2: Leave Roll Number empty 3: Complete the other three fields 4: Click Register	Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll "" (empty), password Str0ng#Pass	Registration rejected; no account created; inline error "Roll number is required"		
UT_003	User Registration	FR-001 — Reject a non-institutional email domain	Registration page reachable; accepted domain configured as pesu.pes.edu	1: Open /register 2: Enter a personal email 3: Complete remaining fields 4: Click Register	Email arjun.rao@gmail.com, other fields valid	Registration rejected; error "Please register with your institutional email address"		
UT_004	User Registration	FR-001 — Reject a malformed email address	Registration page reachable	1: Open /register 2: Enter an email with no @ 3: Complete remaining fields 4: Click Register	Email arjun.rao.pesu.pes.edu, other fields valid	Registration rejected; error "Enter a valid email address"; no account created		
UT_005	User Registration	FR-001 — Reject a duplicate institutional email	STU_A already registered	1: Open /register 2: Enter STU_A's email with a fresh roll number 3: Click Register	Email stu.a@pesu.pes.edu, roll PES1UG24CS999	Registration rejected; error "An account with this email already exists"; STU_A unmodified		
UT_006	User Registration	FR-001 — Reject a duplicate roll number	STU_A already registered	1: Open /register 2: Enter a fresh email with STU_A's roll number 3: Click Register	Email new.student@pesu.pes.edu, roll PES1UG24CS001	Registration rejected; error "This roll number is already registered"		
IT_001	User Registration	FR-001 — Registration persists a Student row with a hashed password	Test database reachable and inspectable	1: Register a new Student 2: Query the users table for that email 3: Inspect the stored password column	Email arjun.rao@pesu.pes.edu, password Str0ng#Pass	Exactly one row exists; role is STUDENT; the password column holds a salted hash; the literal Str0ng#Pass appears nowhere in the database or logs		
IT_002	User Registration	FR-001 — Two concurrent registrations of the same roll number produce one account	Test database reachable; no account for roll PES1UG24CS222	1: Fire two registration requests with the same roll number simultaneously 2: Query the users table	Two requests, both roll PES1UG24CS222, different emails	Exactly one account is created; the other is rejected as a duplicate; no partial or orphaned row		
ST_001	User Registration	FR-001 — End-to-end registration through to a usable account	TEST environment running	1: Open the portal as an anonymous visitor 2: Register with valid details	Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll	Registration succeeds and the new account can immediately sign in and		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3: Sign in with the new credentials 4: Confirm the Student dashboard loads	PES1UG24CS111, password `Str0ng#Pass`	reach the Student dashboard		

UC-02 — Sign in and reach a role-appropriate dashboard

Verifies `FR-002`, `FR-003`, and the sign-in lockout of `NFR-012`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_007	Authentication	FR-002 — Sign in with correct credentials	STU_A exists and is active	1: Open `/login` 2: Enter email and password 3: Click Sign in	`stu.a@pesu.pes.edu` / `Passw0rd!A`	Sign-in succeeds; a session is established; the Student dashboard is displayed		
UT_008	Authentication	FR-002 — Reject a correct email with a wrong password	STU_A exists and is active	1: Open `/login` 2: Enter the correct email and a wrong password 3: Click Sign in	`stu.a@pesu.pes.edu` / `WrongPass1`	Sign-in refused; no session established; message "Invalid email or password" — it does not reveal that the email exists		
UT_009	Authentication	FR-002 — Reject an unknown email without disclosing that it is unknown	No account exists for the email used	1: Open `/login` 2: Enter an unregistered email and any password 3: Click Sign in	`nobody@pesu.pes.edu` / `AnyPass123`	Sign-in refused with the identical message and a comparable response time to UT_008, so accounts cannot be enumerated		
UT_010	Authentication	FR-002 — Reject sign-in with an empty password	STU_A exists	1: Open `/login` 2: Enter a valid email and leave the password empty 3: Click Sign in	`stu.a@pesu.pes.edu` / "" (empty)	Sign-in refused; inline error "Password is required"; no authentication attempt is recorded		
UT_011	Authorization	FR-003 — Every account carries exactly one role	STU_A, FAC_A and ADM_A exist	1: Query the stored record for each of the three accounts 2: Inspect the role field	The three fixture accounts	Each holds exactly one role, drawn from Student, Faculty or Administrator; no account holds none or more than one		
IT_003	Authentication	FR-002 — The session identifier is regenerated on sign-in	STU_A exists; browser developer tools available	1: Load `/login` and record the pre-authentication session cookie 2: Sign in successfully 3: Record the cookie again	`stu.a@pesu.pes.edu` / `Passw0rd!A`	The identifier differs before and after; the pre-authentication identifier is invalidated, so a fixated session cannot be reused		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				and compare				
IT_004	Authorization	FR-003 — A Student is refused a Faculty-only API endpoint called directly	STU_A signed in	1: Sign in as STU_A 2: Using the session token, call POST `/api/questions` directly, bypassing the UI 3: Inspect the response	Valid Student session token; a well-formed question payload	Response is HTTP 403; no question is created; the body carries no Faculty data		
IT_005	Authorization	FR-003 — A Faculty member is refused an Administrator-only endpoint	FAC_A signed in	1: Sign in as FAC_A 2: Call POST `/api/admin/courses` with a valid payload 3: Inspect the response	Valid Faculty session token; a well-formed course payload	Response is HTTP 403; no course is created		
IT_006	Authorization	FR-003 — An unauthenticated client is refused every protected endpoint	Application running; no session held	1: Enumerate the application's endpoints 2: Call each with no session cookie, excluding sign-in and registration 3: Record each status code	The full endpoint list	Every protected endpoint returns HTTP 401; no endpoint defaults to open		
IT_038	Authentication	NFR-012 — Five failed sign-ins within 15 minutes lock the account for 15 minutes	STU_A exists and is active; no failed sign-ins recorded for it	1: Sign in as STU_A with a wrong password five times 2: Sign in with the correct password 3: Advance the recorded failures by 16 minutes 4: Sign in with the correct password again	`stu.a@pesu.pes.edu` / `WrongPass1` ×5, then `Passw0rd!A`	Step 2 is refused with "Too many failed sign-ins. Try again in 15 minutes." even though the password is right, and no session is established; step 4 succeeds		
ST_002	Authentication & Authorization	FR-002, FR-003 — Each role lands on its own dashboard and cannot reach another's	STU_A, FAC_A, ADM_A exist and are active	1: Sign in as STU_A, note the landing page, attempt `/faculty/questions`, sign out 2: Repeat as FAC_A, attempting `/admin/courses` 3: Repeat as ADM_A, confirming the admin dashboard loads	The three fixture accounts	Each role lands on its own dashboard; every cross-role navigation is refused with HTTP 403 and an authorisation message; no other role's data is rendered		

UC-03 — Manage courses and assign faculty

Verifies FR-004.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_012	Course Management	FR-004 — Create a course with code, title and semester	ADM_A signed in	1: Open /admin/courses 2: Click New course 3: Enter code, title and semester 4: Save	Code UE24CS351A, title Database Management Systems, semester 5	The course is created and listed as Active; message "Course created"		
UT_013	Course Management	FR-004 — Reject a course with any mandatory field empty	ADM_A signed in	1: Open the New course form 2: Leave the Title empty 3: Complete code and semester 4: Save	Code UE24CS999A, title "" (empty), semester 5	Creation rejected; inline error "Course title is required"; no course created		
UT_014	Course Management	FR-004 — Reject a duplicate code within the same semester	CRS_X exists as UE24CS351A in semester 5	1: Open the New course form 2: Enter the same code and semester with a different title 3: Save	Code UE24CS351A, title DBMS Repeat, semester 5	Creation rejected; error "A course with this code already exists for the selected semester"		
UT_015	Course Management	FR-004 — Reject assigning a non-Faculty account as Faculty	CRS_X exists; STU_A is a Student	1: Open CRS_X 2: Open the Assign faculty picker 3: Attempt to assign STU_A	STU_A = stu.a@pesu.pes.edu	The assignment is rejected; only Faculty-role accounts are selectable, or the attempt raises "Only Faculty accounts can be assigned to a course"		
UT_016	Course Management	FR-004 — Reject a duplicate faculty assignment	FAC_A already assigned to CRS_X	1: Open CRS_X 2: Assign FAC_A again 3: Inspect the assigned list	FAC_A on CRS_X	The duplicate is rejected or ignored; FAC_A appears exactly once		
IT_007	Course Management	FR-004 — Archiving preserves history while blocking new work	CRS_X has enrolled Students and one completed examination	1: As ADM_A, archive CRS_X 2: Attempt to create a new examination under it 3: Open its past examination's results	CRS_X with EXM_1 completed	New examinations and enrollments are refused with "This course is archived"; the existing examination, attempts and results remain readable		
IT_008	Course Management	FR-004 — Assigning a second faculty grants both authoring rights	CRS_X assigned to FAC_A only	1: As ADM_A, assign FAC_B to CRS_X 2: Sign in as FAC_B 3: Create a question under CRS_X	FAC_B newly assigned to CRS_X	CRS_X appears in FAC_B's course list and the question is created successfully		
ST_003	Course Management	FR-004 — End-to-end course setup by an Administrator	TEST environment running	1: Sign in as ADM_A 2: Create a course 3: Assign FAC_A to it 4: Sign in as FAC_A and confirm the course is listed 5: Archive the course as ADM_A	Code UE24CS360A, title Operating Systems, semester 5; FAC_A	Every stage behaves as specified; the archived course remains readable but closed to new work		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				6: Confirm FAC_A can no longer add examinations to it				

UC-04 — Enroll students and gate examination visibility

Verifies FR-005, FR-006.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_017	Enrollment	FR-005 — Faculty enrolls a Student into their own course	FAC_A signed in and assigned to CRS_X; STU_A exists	1: Open CRS_X 2: Click Enroll student 3: Search for STU_A and confirm	STU_A = roll PES1UG24CS001	STU_A appears on the CRS_X roster; message "Student enrolled"		
UT_018	Enrollment	FR-005 — Reject a duplicate enrollment	STU_A already enrolled in CRS_X	1: Open CRS_X 2: Enroll STU_A again 3: Inspect the roster	STU_A on CRS_X	The duplicate is rejected with "This student is already enrolled"; the roster shows STU_A once		
UT_019	Enrollment	FR-005 — Reject enrolling a Faculty or Administrator account	FAC_A signed in on CRS_X	1: Open the Enroll student picker 2: Search for FAC_B and attempt to enroll	FAC_B = fac.b@pes.edu	The enrollment is rejected; only Student-role accounts are selectable, or the attempt raises "Only Student accounts can be enrolled"		
UT_020	Enrollment	FR-005 — Faculty cannot enroll into a course they are not assigned to	FAC_B assigned to CRS_Y only	1: Sign in as FAC_B 2: Call the enroll endpoint for CRS_X directly	FAC_B, target CRS_X, student STU_A	The call is refused with HTTP 403; no enrollment is created		
UT_021	Enrollment	FR-006 — A Student sees only the examinations of enrolled courses	STU_A enrolled in CRS_X only; both CRS_X and CRS_Y have published examinations	1: Sign in as STU_A 2: Open the dashboard 3: List the examinations shown	STU_A; CRS_X and CRS_Y each with one published exam	Only the CRS_X examination is listed; the CRS_Y examination does not appear		
IT_009	Enrollment	FR-006 — Direct URL access to a non-enrolled examination is refused	STU_A not enrolled in CRS_Y; a CRS_Y examination exists with a known ID	1: Sign in as STU_A 2: Request the CRS_Y examination URL directly 3: Inspect the response	STU_A; CRS_Y examination ID	Response is HTTP 403; no title, instructions or questions appear in the body		
IT_010	Enrollment	FR-006 — The start-attempt endpoint refuses a non-enrolled	STU_A not enrolled in CRS_Y; a live CRS_Y examination exists	1: Sign in as STU_A 2: Call POST /api/attempts for the CRS_Y examination	STU_A; live CRS_Y examination	The call is refused with HTTP 403; no attempt record is created		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		Student		3: Query the attempts table				
IT_011	Enrollment	FR-006 — Unenrolling removes an upcoming examination from the dashboard	STU_A enrolled in CRS_X, which has an upcoming examination	1: Confirm the examination is listed 2: As FAC_A, remove STU_A from CRS_X 3: Refresh the dashboard 4: Attempt to start the examination	STU_A, CRS_X, upcoming EXM_1	The examination disappears from the dashboard and cannot be started		
ST_004	Enrollment	FR-005, FR-006 — End-to-end cohort isolation between two courses	TEST environment; CRS_X and CRS_Y each with a published examination	1: As FAC_A, enroll STU_A into CRS_X 2: As FAC_B, enroll STU_B into CRS_Y 3: Sign in as STU_A and list visible examinations 4: Sign in as STU_B and do the same 5: From each account, request the other's examination URL	STU_A → CRS_X; STU_B → CRS_Y	Each Student sees only their own course's examination; each cross-course URL request is refused with HTTP 403		

UC-05 — Author and manage question bank entries

Verifies FR-007, FR-008, FR-009.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_022	Question Bank	FR-007 — Create a single-correct MCQ with four options	FAC_A signed in and assigned to CRS_X	1: Open /faculty/questions/new 2: Enter the stem 3: Add four options and mark one correct 4: Complete the metadata 5: Save	Stem Which normal form removes transitive dependency?, options 1NF/2NF/3NF/BCNF, correct 3NF	The question is saved to the CRS_X bank and appears in the question list		
UT_023	Question Bank	FR-007 — Reject a question with fewer than two options	FAC_A signed in	1: Open the new question form 2: Enter a stem and a single option 3: Save	Stem valid, one option 3NF	Creation rejected; error "A question must have between 2 and 6 options"		
UT_024	Question Bank	FR-007 — Reject a question with more than six options	FAC_A signed in	1: Open the new question form 2: Enter a stem and seven	Stem valid, seven options A–G	Creation rejected with the same boundary message; no question is created		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				options 3: Save				
UT_025	Question Bank	FR-007 — Reject a question with no option marked correct	FAC_A signed in	1: Enter a stem and four options 2: Mark none as correct 3: Save	Four options, zero marked correct	Creation rejected; error "Exactly one option must be marked correct"		
UT_026	Question Bank	FR-007 — Reject a single-correct question with two options marked correct	FAC_A signed in	1: Enter a stem and four options 2: Mark two as correct 3: Save	Four options, 2NF and 3NF both marked correct	Creation rejected with the same message; exactly one correct option is required		
UT_027	Question Bank	FR-008 — Reject a question missing any mandatory metadata field	FAC_A signed in	1: Enter a valid stem and options 2: Leave Difficulty unset 3: Save; repeat for course, topic, marks and negative marks	Stem and options valid; one metadata field empty each time	Each is rejected with an inline error naming the missing field; no question is created		
IT_012	Question Bank	FR-008 — Reject a difficulty value outside the permitted set	FAC_A signed in; API reachable	1: Submit a question via the API with an invalid difficulty 2: Inspect the response	Difficulty `Impossible`	Rejected with HTTP 400 and "Difficulty must be one of Easy, Medium or Hard"		
IT_013	Question Bank	FR-009 — A question in a published examination cannot be edited in place	`Q_SC1` is attached to `EXM_1`, which is published	1: Sign in as FAC_A 2: Open `Q_SC1` 3: Change the stem 4: Save 5: Query the question table	`Q_SC1`, new stem text	The in-place edit is refused; a **new version** is created instead; the version attached to `EXM_1` is byte-identical to before		
IT_014	Question Bank	FR-009 — A question that a Student has answered cannot be deleted	`Q_SC1` attached to `EXM_1`; STU_A has answered it in a submitted attempt	1: Sign in as FAC_A 2: Attempt to delete `Q_SC1` 3: Inspect the attempt's stored response	`Q_SC1`, STU_A's submitted attempt	Deletion refused with "This question has been attempted and cannot be deleted"; the stored response remains resolvable to the original question text		
ST_005	Question Bank	FR-007, FR-008, FR-009 — End-to-end authoring, tagging and version-freeze	TEST environment; FAC_A assigned to CRS_X	1: Sign in as FAC_A 2: Author three tagged questions 3: Attach them to a new examination and publish it 4: Attempt to edit one of the three 5: Confirm a new version is created and the published paper is unchanged	Three MCQs across two topics and two difficulties	Authoring and tagging succeed; once published, the paper's questions are frozen — edits produce versions rather than mutations		

UC-06 — Assemble and publish an examination

Verifies FR-010, FR-011, FR-012.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_028	Exam Authoring	FR-010 — Create an examination with all mandatory attributes	FAC_A signed in and assigned to CRS_X	1: Open /faculty/exams/new 2: Enter title, course, duration, declared total marks and instructions 3: Save	Title DBMS Unit Test 1, course CRS_X, duration 60, total 50, instructions free text	The examination is created in **Draft** state and appears in the Faculty examination list		
UT_029	Exam Authoring	FR-010 — Reject a non-positive duration	FAC_A signed in	1: Open the new examination form 2: Enter a duration of zero 3: Save; repeat with a negative value	Duration 0, then -30	Both rejected with "Duration must be a positive number of minutes"; no examination created		
UT_030	Exam Authoring	FR-010 — Per-examination policy values persist and are read back	A Draft examination exists	1: Open its settings 2: Enable negative marking at 0.5, enable question and option shuffling, set pass mark 40% 3: Save and reload the page	Negative 0.5, shuffle both on, pass mark 40	All policy values persist exactly as set and are shown unchanged after reload		
UT_031	Exam Authoring	FR-010 — Reject a pass mark outside 0–100	A Draft examination exists	1: Open its settings 2: Enter a pass mark of 150 3: Save; repeat with −10	Pass mark 150, then -10	Both rejected with "Pass mark must be between 0 and 100"; the stored value is unchanged		
UT_032	Exam Authoring	FR-012 — A Draft examination is invisible to Students	A Draft examination exists on CRS_X; STU_A enrolled in CRS_X	1: Sign in as STU_A 2: Open the dashboard 3: Request the Draft examination's URL directly	Draft examination ID	The examination does not appear on the dashboard, and the direct request is refused with HTTP 403 or 404		
UT_033	Exam Authoring	FR-012 — Publishing moves the examination out of Draft state	A Draft examination with questions attached	1: Open the Draft examination 2: Click Publish 3: Inspect the stored state	Draft examination with 25 questions	The state changes from DRAFT to PUBLISHED; the transition is recorded with a server timestamp		
IT_015	Exam Authoring	FR-011 — Questions are attached from the bank and stored against the examination	FAC_A signed in; CRS_X bank holds at least ten questions	1: Open a Draft examination 2: Select five questions from the bank 3: Save 4: Query the examination-question join table	Five known question IDs	Exactly the five selected questions are attached, in the chosen order, with no duplicates and no unselected question present		
IT_016	Exam Authoring	FR-012 — Publishing makes the examination visible to enrolled	A Draft examination on CRS_X; STU_A enrolled, STU_B not	1: As FAC_A, publish the examination 2: Sign in as STU_A and check	Published examination on CRS_X	STU_A sees it listed; STU_B does not, since visibility requires both		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		Students only		the dashboard 3: Sign in as STU_B and check		publication and enrollment		
ST_006	Exam Authoring	FR-010, FR-011, FR-012 — End-to-end paper assembly and publication	TEST environment; CRS_X bank populated; FAC_A assigned	1: Sign in as FAC_A 2: Create a Draft with title, duration and policy values 3: Attach 25 questions from the bank 4: Confirm a Student cannot see it 5: Publish 6: Sign in as an enrolled Student and confirm the paper is listed	60-minute paper, 25 questions, negative `0.5`, pass mark `40`	The paper is invisible while Draft, becomes visible on publication, and carries the configured policy values through unchanged		

UC-07 — Schedule an examination and surface it on the Student dashboard

Verifies `FR-013`, `FR-014`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_034	Scheduling	FR-013 — Schedule a published examination with a valid window	A published examination of 60 minutes' duration exists	1: Open the examination 2: Set window start and end date-times 3: Save	Start `2026-09-10 10:00`, end `2026-09-10 12:00`	The schedule is saved and displayed; message "Examination scheduled"		
UT_035	Scheduling	FR-013 — Reject a window whose end precedes its start	A published examination exists	1: Set an end date-time earlier than the start 2: Save	Start `2026-09-10 12:00`, end `2026-09-10 10:00`	Rejected with "The window end must be later than the window start"; no schedule saved		
UT_036	Scheduling	FR-014 — The dashboard classifies an examination as Upcoming before its window	STU_A enrolled in CRS_X; a CRS_X examination scheduled to start tomorrow	1: Sign in as STU_A 2: Open the dashboard 3: Locate the examination	Window starting `now + 24 h`	Listed under **Upcoming** with its start date-time; no Start button is offered		
UT_037	Scheduling	FR-014 — The dashboard classifies an examination as Live inside its window	STU_A enrolled; a CRS_X examination whose window is currently open	1: Sign in as STU_A 2: Open the dashboard	Window `now − 10 min` to `now + 50 min`	Listed under **Live** with an active Start button		
UT_038	Scheduling	FR-014 — The dashboard classifies a submitted examination as Completed	STU_A has one submitted attempt on `EXM_1`	1: Sign in as STU_A 2: Open the dashboard	`EXM_1` with a submitted attempt	Listed under **Completed**; no Start button is offered		
IT_017	Scheduling	FR-013 — The stored window is timezone-	A published examination; database inspectable	1: Schedule a window from the UI	Start `2026-09-10 10:00 IST`	The stored values are unambiguous (UTC or with		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		unambiguous and server-authoritative		2: Query the stored start and end values 3: Change the client machine's timezone and reload		an explicit offset); the displayed window is unchanged by the client timezone switch		
IT_018	Scheduling	FR-014 — Dashboard classification is driven by server time, not the client clock	STU_A enrolled; a CRS_X examination whose window opens in one hour	1: Sign in as STU_A 2: Advance the client clock by two hours 3: Reload the dashboard	Window opening at `server now + 1 h`; client clock `+2 h`	The examination remains under **Upcoming**; the tampered client clock does not promote it to Live		
ST_007	Scheduling	FR-013, FR-014 — End-to-end scheduling and dashboard transitions	TEST environment; a published 30-minute examination; STU_A enrolled	1: As FAC_A, set a 60-minute window opening in two minutes 2: As STU_A, confirm it shows Upcoming 3: Wait for the window to open and refresh 4: Sit and submit the paper 5: Refresh the dashboard	Duration `30`, window `60 min`	The examination moves Upcoming → Live → Completed on the Student dashboard as server time advances		

UC-08 — Start an attempt and receive the paper

Verifies FR-015, FR-016, FR-017, FR-018.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_039	Exam Delivery	FR-015 — Starting is refused before the window opens	STU_A enrolled; EXM_1 window opens in ten minutes	1: Sign in as STU_A 2: Attempt to start EXM_1	Server time `window start − 10 min`	Refused with "This examination has not opened yet"; no attempt record is created		
UT_040	Exam Delivery	FR-015 — Starting is refused after the window closes	STU_A enrolled; EXM_1 window closed one minute ago	1: Sign in as STU_A 2: Attempt to start EXM_1	Server time `window end + 1 min`	Refused with "This examination has closed"; no attempt record is created		
UT_041	Exam Delivery	FR-015 — Starting is refused when a submitted attempt already exists	STU_A has one submitted attempt on EXM_1; the window is still open	1: Sign in as STU_A 2: Attempt to start EXM_1 again	STU_A, EXM_1, one submitted attempt	Refused with "You have already submitted this examination"; no second attempt is created		
UT_042	Exam Delivery	FR-017 — The countdown is computed from server time, not the client clock	STU_A has an attempt in progress on EXM_1, 60-minute duration, 10 minutes elapsed	1: Note the displayed remaining time 2: Set the client clock back by 30 minutes	Elapsed `10 min`; client clock moved `−30 min`	Remaining time reads approximately 50 minutes both times; the client clock change gains the candidate		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3: Refresh the attempt page 4: Note the remaining time again		nothing		
UT_043	Exam Delivery	FR-018 — The palette shows the correct state for each question	STU_A has an attempt in progress on a 25-question paper	1: Start the attempt and inspect question 5's palette state 2: Open question 5, answer nothing, move on, inspect again 3: Return and select an option, inspect again 4: Mark it for review and inspect again	Question 5 across four interactions	The state moves **Not Visited → Not Answered → Answered → Marked for Review**, matching each interaction exactly		
UT_044	Exam Delivery	FR-018 — Exactly one question is displayed at a time	STU_A has an attempt in progress on a 25-question paper	1: Load the attempt screen 2: Count the question stems rendered in the DOM	25-question paper	Exactly one stem is rendered; the other 24 are not present in the delivered markup		
IT_019	Exam Delivery	FR-016 — The attempt record is created and server-stamped before the paper is delivered	STU_A enrolled; EXM_1 window open; network trace available	1: Click Start 2: Capture the request/response sequence 3: Query the attempts table	STU_A (`stu.a@pesu.pes.edu` / `Passw0rd!A`, enrolled in CRS_X); EXM_1 ("DBMS Unit Test 1", 60 min, 25 questions, published, window open now)	The attempt row exists with a server-generated start timestamp **before** the question payload is returned; no question data is sent if the row cannot be created		
IT_020	Exam Delivery	FR-016 — The stored start time is the server's, not the client's	STU_A enrolled; EXM_1 window open	1: Set the client clock forward by one hour 2: Start the attempt 3: Query the stored start timestamp	Client clock +1 h	The stored start time reflects true server time; the tampered client clock is ignored		
IT_021	Exam Delivery	FR-018 — Palette state survives navigation and reload	STU_A has an attempt with a mix of palette states across ten questions	1: Produce all four states across ten questions 2: Reload the browser 3: Compare every palette state before and after	Ten questions across four states	Every state is identical after reload, because palette state is derived from server-held data rather than browser memory		
ST_008	Exam Delivery	FR-015, FR-016, FR-017, FR-018 — End-to-end paper delivery under a live window	TEST environment; EXM_1 live; STU_A enrolled	1: Sign in as STU_A 2: Confirm starting is refused before the window opens 3: Start once the window is open 4: Confirm the timer begins at the full duration and counts down	60-minute paper, 25 questions	The attempt starts only inside the window, the server-driven timer counts down correctly, one question shows at a time, and the palette reflects every state transition accurately		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				5: Navigate across several questions using the palette 6: Submit				

UC-09 — Autosave and resume after interruption

Verifies FR-019, FR-021.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_045	Attempt Responses	FR-019 — A response is persisted without any explicit save action	STU_A has an attempt in progress; network trace available	1: Select an option on question 4 2: Take no further action 3: Watch the network trace	Question 4, option A	A persistence request is sent automatically within 5 seconds of the selection; no Save button is required or present		
UT_046	Attempt Responses	FR-019 — Changing a response persists the new value, not the old	STU_A has answered question 3 with option B	1: Return to question 3 2: Select option D 3: Wait six seconds 4: Query the stored response	Question 3, B replaced by D	The stored response reads option D; exactly one response is held for the question		
UT_047	Attempt Responses	FR-019 — Clearing a response persists the cleared state	STU_A has answered question 3	1: Return to question 3 2: Click Clear response 3: Wait six seconds 4: Query the stored response	Question 3, cleared	The stored response is empty and the palette returns the question to **Not Answered**		
UT_048	Attempt Responses	FR-021 — Remaining time on resume is computed from the original start	STU_A started EXM_1 at server time T; 60-minute duration	1: Note the start time 2: Wait 20 minutes 3: Reload the attempt 4: Read the remaining time	Start T, reload at T + 20 min	Remaining time reads approximately 40 minutes — derived from the original start, not reset by the reload		
IT_022	Attempt Responses	FR-019 — The autosave reaches the server inside the 5-second budget	STU_A has an attempt in progress; server logs timestamped	1: Select an option and record the client-side timestamp 2: Locate the corresponding server-side write 3: Compute the difference 4: Repeat for ten selections	Ten selections across ten questions	Every response is committed server-side within 5 seconds of selection; the worst observed case is recorded		
IT_023	Attempt Responses	FR-019 — A browser crash loses no response already autosaved	STU_A has an attempt in progress with six questions answered	1: Answer six questions 2: Wait six seconds 3: Kill the browser process without submitting 4: Reopen, sign in and resume	Six answered questions	All six responses are restored; nothing is lost, because persistence did not depend on submission		
IT_024	Attempt	FR-021 — Resuming	STU_A started EXM_1 at	1: Answer four questions	Start T, interruption	All four answers are restored;		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
	Responses	restores responses and does not grant extra time	T; answered four questions by T + 20 min	2: Disconnect the network for five minutes 3: Reconnect and reopen the attempt 4: Inspect the restored answers and remaining time	T+20 to T+25	remaining time reads approximately 35 minutes, so the interruption grants no extra time		
IT_025	Attempt Responses	FR-021 — Resuming on a different device restores the same attempt state	STU_A has an attempt in progress on profile A with three answers	1: Answer three questions on profile A 2: Sign in as STU_A on profile B 3: Open the attempt	Three answered questions; two profiles	Profile B resumes the same attempt with the three answers present and time still measured from the original start		
ST_009	Attempt Responses	FR-019, FR-021 — End-to-end answer, disconnect, resume and submit	TEST environment; EXM_1 live; STU_A enrolled	1: Start EXM_1 2: Answer five questions, changing two and clearing one 3: Disconnect for three minutes 4: Reconnect and resume 5: Verify all responses and remaining time 6: Answer the rest and submit 7: Inspect the stored attempt	25-question paper; five answered, two changed, one cleared	Every selection, change and clear survives the interruption exactly; remaining time reflects the original start; the submitted attempt records precisely the final state of every question		

UC-10 — Automatic submission on timer expiry

Verifies FR-020.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_049	Auto-Submission	FR-020 — The attempt submits itself when remaining time reaches zero	STU_A has an attempt in progress with 30 seconds remaining	1: Answer one more question 2: Take no action and let the timer run to zero 3: Observe the screen	Remaining 30 s, then 0	At zero the attempt is submitted automatically without any click; the Student sees "Time is up. Your paper has been submitted."		
UT_050	Auto-Submission	FR-020 — Auto-submission retains every response persisted up to that instant	STU_A has an attempt with eight answered questions and 20 seconds remaining	1: Confirm eight answers 2: Let the timer expire 3: Query the submitted attempt's responses	Eight answered of 25	All eight responses are present; the remaining 17 are recorded as unattempted		
UT_051	Auto-Submission	FR-020 — No further response is accepted after auto-submission	STU_A's attempt has just auto-submitted	1: Attempt to select an option on the last displayed question	Auto-submitted attempt ID	The UI is read-only; the direct API call is refused with "This attempt has		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				2: Call the response-save endpoint directly with the old attempt ID		already been submitted"; the stored responses are unchanged		
UT_052	Auto-Submission	FR-020 — The attempt state becomes Submitted with a server timestamp	An attempt that has just auto-submitted	1: Query the attempt record 2: Inspect its state and submission timestamp	Auto-submitted attempt	State reads SUBMITTED; the submission timestamp is server-generated and equals the attempt start plus the configured duration		
IT_026	Auto-Submission	FR-020 — Expiry is enforced server-side even if the browser is closed	STU_A has an attempt in progress with one minute remaining	1: Answer two questions 2: Close the browser entirely 3: Wait until after the duration elapses 4: Sign in again and open the examination 5: Query the attempt record	Browser closed at T − 60 s	The attempt is recorded as submitted with the two answers retained; expiry did not depend on the browser being open		
IT_027	Auto-Submission	FR-020 — A client-side clock change cannot defer auto-submission	STU_A has an attempt with two minutes remaining	1: Set the client clock back by one hour 2: Let real time pass beyond the true expiry 3: Attempt to answer a question	Client clock −1 h; real elapsed time beyond duration	The attempt is auto-submitted at the true server expiry; further answers are refused		
IT_028	Auto-Submission	FR-020, FR-022 — An auto-submitted attempt enters evaluation like a manual one	Auto-submission has just occurred for STU_A on EXM_1	1: Let the attempt auto-submit 2: Query the evaluation record within 5 seconds	Auto-submitted attempt on a 25-question paper	The attempt is evaluated automatically with a total score, percentage and Pass/Fail outcome, identical in treatment to a manually submitted attempt		
ST_010	Auto-Submission	FR-020 — End-to-end timed expiry on a short paper	TEST environment; a 3-minute examination with five questions; STU_A enrolled	1: Start the examination 2: Answer three of the five questions 3: Let the full three minutes elapse without submitting 4: Observe the screen at zero 5: Sign in as FAC_A and inspect the attempt	3-minute paper, five questions, three answered	The paper submits itself exactly at expiry, the three answers are retained, the two unanswered are recorded as unattempted, and Faculty sees a normally submitted attempt		

UC-11 — Evaluate a submitted attempt automatically

Verifies FR-022, FR-023, FR-024.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_053	Evaluation	FR-023 — Award full positive marks for a correct response	Scoring engine callable in isolation; Q_SC1 = +2 / −0.5, correct option B	1: Invoke the evaluator with a response of option B 2: Read the awarded marks	Response B, correct B, positive 2, negative 0.5	The question is awarded exactly +2.0		
UT_054	Evaluation	FR-023 — Deduct the configured negative marks for an incorrect response	Scoring engine callable; negative marking enabled at 0.5	1: Invoke the evaluator with a response of option A 2: Read the awarded marks	Response A, correct B, positive 2, negative 0.5	The question is awarded exactly −0.5		
UT_055	Evaluation	FR-023 — Award zero and apply no deduction to an unattempted question	Scoring engine callable; negative marking enabled at 0.5	1: Invoke the evaluator with no response recorded 2: Read the awarded marks	Response null, correct B, negative 0.5	The question is awarded exactly 0.0; the penalty is **not** applied — the boundary that separates skipping from guessing		
UT_056	Evaluation	FR-023 — Apply no deduction when negative marking is disabled	Scoring engine callable; negative marking disabled on the examination	1: Invoke the evaluator with an incorrect response 2: Read the awarded marks	Response A, correct B, negative marking off	The question is awarded exactly 0.0, not a negative value		
UT_057	Evaluation	FR-024 — Compute total, percentage and a Pass outcome at the pass mark	Scoring engine callable; pass mark 40%	1: Evaluate an attempt scoring 20 of 50 2: Read total, percentage and outcome	Total 20, declared total 50, pass mark 40%	Total 20, percentage 40.0%, outcome **Pass** — the boundary is inclusive		
UT_058	Evaluation	FR-024 — Compute a Fail outcome just below the pass mark	Scoring engine callable; pass mark 40%	1: Evaluate an attempt scoring 19.5 of 50 2: Read the outcome	Total 19.5, declared total 50, pass mark 40%	Percentage 39.0%, outcome **Fail**		
IT_029	Evaluation	FR-022 — Every submitted attempt is evaluated without human intervention	EXM_1 with three submitted attempts	1: Submit three attempts 2: Take no Faculty action of any kind 3: Query the evaluation records	Three submitted attempts	All three carry a computed score, percentage and outcome; none awaits a manual trigger or approval		
IT_030	Evaluation	FR-022, NFR-002 — A 100-question attempt is evaluated within 5 seconds	An examination of 100 questions with a submitted attempt; server timestamps available	1: Submit and record the server submission timestamp 2: Poll for the evaluation record and record its completion timestamp 3: Compute the elapsed time 4: Repeat over ten attempts	100-question paper, ten attempts	Every evaluation completes within 5 seconds of submission; the worst observed elapsed time is recorded		
IT_031	Evaluation	FR-023, FR-024 — A mixed attempt is scored correctly end to	An examination of 10 questions at +2 / −0.5 with a submitted attempt	1: Submit an attempt with 6 correct, 3 incorrect and 1 unattempted	6 correct, 3 incorrect, 1 skipped; +2 / −0.5; declared total 20;	Total is (6 × 2) − (3 × 0.5) − 0 = 10.5; percentage 52.5%; outcome		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		end		2: Query the computed total, percentage and outcome	pass mark 40%	**Pass**; the skipped question contributes exactly zero		
ST_011	Evaluation	FR-022, FR-023, FR-024 — End-to-end automatic scoring of a full cohort	TEST environment; EXM_1 with 25 questions; five enrolled Students	1: Have five Students sit and submit EXM_1 with differing answer patterns 2: Take no Faculty action 3: Inspect each attempt's score, percentage and outcome 4: Hand-compute two of the five from the answer key and compare	Five attempts across a 25-question paper	All five are evaluated automatically; the two hand-computed totals match the system's exactly; Pass/Fail is correct against the configured pass mark		

UC-12 — Publish results and display the scorecard

Verifies FR-025, FR-026, FR-027.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_059	Results	FR-025 — An evaluated result is withheld until Faculty publishes	STU_A has a submitted, evaluated attempt on EXM_1; results not published	1: Sign in as STU_A 2: Open the dashboard and the examination 3: Look for a score	Evaluated but unpublished result	No score, percentage or outcome is shown; the examination reads "Results not yet published"		
UT_060	Results	FR-026 — Publishing makes the result visible to the Student	The situation of UT_059	1: Sign in as FAC_A 2: Publish the results of EXM_1 3: Sign in as STU_A and reopen the examination	EXM_1 results published	The scorecard is now visible with total, percentage and outcome		
UT_061	Results	FR-026 — Unpublishing withdraws a previously visible result	EXM_1 results published and seen by STU_A	1: Sign in as FAC_A 2: Unpublish the results 3: Sign in as STU_A and reopen the examination	EXM_1 results unpublished	The scorecard is no longer shown; the examination reverts to "Results not yet published"; the stored score is unchanged		
UT_062	Results	FR-027 — The scorecard shows total, percentage, outcome and per-question marks	EXM_1 results published; STU_A scored 32 of 50	1: Sign in as STU_A 2: Open the scorecard 3: Inspect every displayed field	Total 32, declared 50, pass mark 40%, 25 questions	Total 32, percentage 64.0%, outcome **Pass**, and a per-question line for all 25 questions showing the marks obtained on each		
UT_063	Results	FR-027 — Per-question marks on the scorecard sum to the reported total	EXM_1 results published for STU_A	1: Open the scorecard 2: Sum the per-question marks 3: Compare with the displayed	25 per-question values including negatives	The sum equals the displayed total exactly		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				total				
IT_032	Results	FR-025 — An unpublished result is unreachable through the API, not merely hidden in the UI	STU_A has an evaluated but unpublished result	1: Sign in as STU_A 2: Call GET `/api/results/{attemptId}` directly 3: Inspect the response body	STU_A's attempt ID; results unpublished	Refused with HTTP 403; no score, percentage or per-question mark appears anywhere in the body		
IT_033	Results	FR-025 — One Student cannot read another's result	EXM_1 results published; STU_A and STU_B both attempted	1: Sign in as STU_A 2: Call the result endpoint with STU_B's attempt ID	STU_B's attempt ID	Refused with HTTP 403; STU_B's score is not disclosed		
IT_034	Results	FR-026, FR-027 — The published scorecard matches the stored evaluation exactly	EXM_1 sat by five Students and published	1: Publish results 2: For each Student, open the scorecard 3: Compare each displayed total, percentage and outcome against the stored evaluation record	Five evaluated attempts	Every displayed figure matches its stored evaluation exactly; no rounding or transcription discrepancy		
ST_012	Results	FR-025, FR-026, FR-027 — End-to-end publication and scorecard	TEST environment; EXM_1 sat by five Students	1: Confirm no Student can see a score before publication 2: As FAC_A, publish 3: As each Student, open the scorecard and verify the figures 4: Unpublish and confirm the scorecards disappear	Five evaluated attempts	Results stay hidden until published, every scorecard matches the stored evaluation, and unpublishing withdraws visibility without altering the data		

UC-13 — Log integrity events and maintain the audit trail

Verifies FR-028, FR-029, FR-030.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_064	Integrity	FR-028 — A single focus-loss event is recorded with a timestamp	STU_A has an attempt in progress on EXM_1	1: Switch to another browser tab once 2: Return to the attempt 3: Query the integrity event log for the attempt	One tab switch	Exactly one focus-loss event is recorded against the attempt, carrying a server-side timestamp		
UT_065	Integrity	FR-028 — Repeated focus losses are each recorded separately	STU_A has an attempt in progress	1: Switch away and back three times 2: Query the event log	Three tab switches	Exactly three events are recorded, each with its own distinct timestamp; none is coalesced or dropped		
UT_066	Integrity	FR-028 — No event is	STU_A has an attempt in	1: Complete the attempt without	Zero tab switches	The event log for the		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		recorded when focus is never lost	progress	leaving the window 2: Query the event log		attempt is empty; the detector does not fire on ordinary in-page interaction		
UT_067	Integrity	FR-028 — The recorded timestamp is server-side, not client-supplied	STU_A has an attempt in progress	1: Set the client clock forward by one hour 2: Switch tabs once 3: Query the recorded event timestamp	Client clock +1 h; one tab switch	The stored timestamp reflects true server time, not the tampered client clock		
UT_068	Audit	FR-030 — An audit entry records actor, action, entity, timestamp and changed values	FAC_A signed in; audit log inspectable	1: Edit a question's stem 2: Query the audit log for that question	Question Q_SC1, stem changed	One entry exists carrying the actor FAC_A, the action, the affected entity, a server timestamp, and both the previous and the new stem values		
IT_035	Integrity	FR-029 — No correct-answer data reaches the candidate during an active attempt	STU_A has an attempt in progress on EXM_1	1: Capture every HTTP response delivered to the browser during the attempt 2: Search the payloads for `isCorrect`, `correctOption`, `answerKey` and the known correct values 3: Inspect the rendered HTML and client-side JavaScript state	EXM_1; known answer key	No response body, HTML attribute or client-side variable exposes which option is correct; the key is absent from everything the browser receives		
IT_036	Audit	FR-030 — Every mutation class writes an audit entry	ADM_A and FAC_A signed in	1: Create a user, a course, a question and an examination, then publish a result 2: Query the audit log for each	One mutation of each of the five entity classes	An entry exists for every one of the five; no mutation class is silently unlogged		
IT_037	Audit	FR-030 — The audit log is append-only and resists modification	Audit entries exist; database reachable	1: Attempt to update an existing audit row through the application 2: Attempt to delete one 3: Inspect the table's constraints	An existing audit entry	Both attempts are refused; the application exposes no update or delete path for audit rows; the entry is unchanged		
ST_013	Integrity & Audit	FR-028, FR-029, FR-030 — End-to-end evidentiary trail across an examination cycle	TEST environment; EXM_1 live; STU_A enrolled	1: Sit EXM_1 as STU_A, switching tabs twice 2: Confirm no answer key appears in any captured payload 3: Submit 4: As FAC_A, publish results 5: As ADM_A, open the audit log filtered to EXM_1 and reconcile it against everything that occurred	One attempt, two focus-loss events, one publication	The log accounts for every state change — attempt creation, submission, evaluation, publication — the two integrity events are attached to the attempt for Faculty review, and no answer key ever reached the candidate		

7. Coverage summary

Use case	Requirements verified	Unit	Integration	System	Total
UC-01 Register a Student account	FR-001	6	2	1	9
UC-02 Sign in and role-based access	FR-002, FR-003, NFR-012	5	5	1	11
UC-03 Manage courses and assign faculty	FR-004	5	2	1	8
UC-04 Enroll students and gate visibility	FR-005, FR-006	5	3	1	9
UC-05 Author question bank entries	FR-007, FR-008, FR-009	6	3	1	10
UC-06 Assemble and publish an examination	FR-010, FR-011, FR-012	6	2	1	9
UC-07 Schedule and surface on the dashboard	FR-013, FR-014	5	2	1	8
UC-08 Start an attempt and receive the paper	FR-015, FR-016, FR-017, FR-018	6	3	1	10
UC-09 Autosave and resume	FR-019, FR-021	4	4	1	9
UC-10 Automatic submission on expiry	FR-020	4	3	1	8
UC-11 Evaluate automatically	FR-022, FR-023, FR-024	6	3	1	10
UC-12 Publish results and scorecard	FR-025, FR-026, FR-027	5	3	1	9
UC-13 Integrity events and audit trail	FR-028, FR-029, FR-030	5	3	1	9
Total	**30 of 30**	**68**	**38**	**13**	**119**

Every use case carries at least eight test cases, and every one of the 30 functional requirements of `Documents/SRS.pdf` is exercised by at least one test case — satisfying NFR-007.

8. Open items affecting these test cases

Two decisions remain outstanding with the course instructor, and each leaves specific expected values provisional:

Open question	Test cases affected	Effect if resolved differently
Negative-marking convention — a flat deduction per wrong answer, or a fraction of the question's marks	UT_054, UT_056, IT_031	The expected values assume a **flat** deduction of 0.5. A fractional convention changes each expected total
Whether a Pass at exactly the pass mark is inclusive	UT_057	The expected outcome assumes **inclusive** — 40.0% against a 40% pass mark is a Pass